Lesson 2: Key Press Reactor

GDevelop

VIDEO GAME CREATION

This week: sequences, events, and the game loop. Your job is to build something that reacts when you press a key - and this is also the start of your Semester 1 project, so save it somewhere safe and give it a name you'll recognize.

Tip: this lesson's Assets folder already has a ready-to-use placeholder character image (and an optional sound effect for Level 3) - you don't need to find or make your own art today.

Setup

1. Create a new blank project named [YourName]Game.
2. Add a Sprite object using one of the placeholder images from this lesson's Assets folder (e.g. player_circle_blue.png), name it Player.

Level 1 - Minimal (MVP)

One key press causes one visible reaction. This is a complete finish for today - reaching this is enough to move on to next week.

3. Select Player, click "Add a behavior," choose "Top-down movement," click Apply.
4. Click the green Preview button and test with the arrow keys.
5. (Alternative manual method: add an event with condition "Key pressed [Right]" and action "Change the object's X position.")

Level 2 - Complete (Alpha/Beta)

All 4 arrow keys move your character in 4 directions.

- With the Top-down movement behavior, all 4 arrow keys already work - confirm by testing all 4 directions in Preview. If you used the manual method for Level 1, add the other 3 direction events now.

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Animate: make Player blink while moving, and stay solid (visible) when idle. Flash visibility (blink) isn't a standalone action sitting in a general list - it belongs to a behavior, the same way Angle of movement (used in the Recolor goal below) belongs to the Top-down movement behavior. Select Player, open its Behaviors tab, click Add a behavior, search "Flash," and Apply - only once that's attached will "Flash visibility (blink)" (start) and "Stop flashing visibility (blink)" (stop) show up when you search Player's actions.

- Sound: play sfx_blip.wav in a beep-beep pattern while moving, silent when idle - use "Play a sound" together with a scene timer so it retriggers instead of overlapping itself every frame.
- Recolor: give Player a different Animation per direction rather than tinting one. Add the other 4 placeholder circles as extra Animations on the Player object (Edit object > Animations tab > Add an animation), naming them Red/Blue/Green/Yellow, plus your default one renamed Orange. The thing you're building here is 5 separate Events, not conditions/actions on their own - each Event is one row that bundles together its own Condition(s) and Action(s), and an Event's Action(s) only run when all of that SAME Event's Condition(s) are true. There's no separate step to "connect" a condition to an action; they're never independent pieces to begin with, only ever parts of the one Event that owns them. Build the first one: click Add new event, then, inside that Event, click its own Add condition (small text, top-left of the row), pick Player, then look for a separate Top-down movement category in the list (not Player's own conditions) and choose Angle of movement - set it to 0 degrees with a Tolerance of about 45. Still inside that same Event, click its Add action, pick Player again, and in the action search list look for Animation (by name) under Animations and images - it's labeled that way in the picker, but displays as "Change the animation of Player..." once you've placed it - set it to "Red". Click Add new event again and repeat for the other 3 directions (90/180/270 degrees for Down/Left/Up), plus a fifth new Event using an "Is moving" = False condition instead, with its action set to "Orange" for idle.
- Clamp: keep Player on screen by adding the Stay On Screen (2D) behavior - zero events required.
- Structure: keep all four of the above together in one labeled event group - not scattered across separate, unlabeled events.

If Something's Wrong

- Preview doesn't launch: check the log panel at the bottom of the window for a red error.
- Object doesn't move: confirm the behavior was actually added (check the object's Behaviors tab), and that you're testing in Preview, not the static scene editor.
- Recolor conditions/actions ended up in the wrong Event, or all crammed into one: no need to delete and rebuild - click Add new event to create an empty one, then drag and drop the misplaced Condition(s)/Action(s) into it (or out of it) to sort them into the right Event.
- Can't find Flash visibility (blink), Angle of movement, or another action/condition that should exist: some actions and conditions belong to a behavior rather than to the object itself, so they only appear once that behavior has been added - check the object's Behaviors tab first (Add a behavior), then search the action/condition list again.

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